201317 201314 201320 201417 201415 201419 201329 201327 201332 201428
201425 201430

Homework 1

Jaisel Singh (js6897) October 15, 2026

Question 1: A First Look at Denoising Models

(a) Posterior. Both $p(x_0)$ and $p(x_1 \mid x_0)$ are Gaussian with mean linear in x_0, so (x_0, x_1) is jointly Gaussian and the posterior is Gaussian before any algebra; the work is to find its two parameters. By Bayes' rule, keeping only the factors that involve x_0,

$$p(x_0 \mid x_1) \;\propto\; p(x_0)\, p(x_1 \mid x_0) \;\propto\; \exp\left\{-\frac{x_0^2}{2\tau_0^2} - \frac{(x_1 - \sqrt{1-\beta}\, x_0)^2}{2\beta}\right\}.$$

Expanding the square and dropping $x_1^2/(2\beta)$, which is constant in x_0, the exponent becomes

$$-\frac{x_0^2}{2\tau_0^2} - \frac{(1-\beta)\, x_0^2 - 2\sqrt{1-\beta}\, x_0 x_1}{2\beta} \;=\; -\frac{1}{2}\underbrace{\left(\frac{1}{\tau_0^2} + \frac{1-\beta}{\beta}\right)}_{\lambda} x_0^2 \;+\; \underbrace{\frac{\sqrt{1-\beta}\, x_1}{\beta}}_{h}\, x_0.$$

Completing the square, any density proportional to $\exp\{-\frac{1}{2}\lambda x_0^2 + h x_0\}$ is $\mathcal{N}(h/\lambda,\, 1/\lambda)$. Here $\lambda = \big(\beta + (1-\beta)\tau_0^2\big)/(\beta\tau_0^2)$, so

$$\frac{1}{\lambda} = \frac{\beta\,\tau_0^2}{\beta + (1-\beta)\tau_0^2}, \qquad \frac{h}{\lambda} = \frac{\sqrt{1-\beta}\, x_1}{\beta} \cdot \frac{\beta\,\tau_0^2}{\beta + (1-\beta)\tau_0^2} = \frac{\sqrt{1-\beta}\,\tau_0^2\, x_1}{\beta + (1-\beta)\tau_0^2},$$

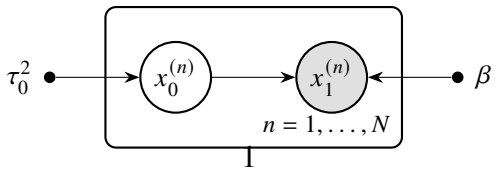
$$\boxed{x_0 \mid x_1 \;\sim\; \mathcal{N}\left(\frac{\sqrt{1-\beta}\,\tau_0^2}{\beta + (1-\beta)\tau_0^2}\, x_1,\;\; \frac{\beta\,\tau_0^2}{\beta + (1-\beta)\tau_0^2}\right).}$$

Check. Jointly $\mathbb{E}[x_1] = 0$, $\mathrm{Var}(x_1) = (1-\beta)\tau_0^2 + \beta$ by total variance, and $\mathrm{Cov}(x_0, x_1) = \mathbb{E}[x_0\, \mathbb{E}(x_1 \mid x_0)] = \sqrt{1-\beta}\,\tau_0^2$; Gaussian conditioning, mean $\mathrm{Cov}(x_0, x_1)\, x_1/\mathrm{Var}(x_1)$ and variance $\tau_0^2 - \mathrm{Cov}(x_0, x_1)^2/\mathrm{Var}(x_1)$, reproduces both boxed expressions.

The mean is $x_1/\sqrt{1-\beta}$, the inversion of the deterministic scaling, times $(1-\beta)\tau_0^2/\big((1-\beta)\tau_0^2 + \beta\big)$, the share of $\mathrm{Var}(x_1)$ due to signal rather than injected noise; the posterior therefore shrinks that inversion toward the prior mean 0. Its variance is free of x_1: an observation relocates the posterior without changing its spread.

(b) Limits. As $\beta \to 0$ the mean tends to x_1 and the variance to 0: the step is the identity, so x_1 pins down x_0 exactly and the posterior is a point mass. As $\beta \to 1$ the mean tends to 0 and the variance to τ_0^2: x_1 is pure noise, the likelihood is flat, and the posterior falls back on the prior $\mathcal{N}(0, \tau_0^2)$. Since $d\sigma^2/d\beta = \tau_0^4/\big(\beta + (1-\beta)\tau_0^2\big)^2 > 0$, the variance climbs monotonically between these limits, so β sets how much of x_0 survives the step.

(c) Graphical model. The statement describes one (x_0, x_1) pair; the plate shows the i.i.d. replication over N of them. Shaded $x_1^{(n)}$ is observed, $x_0^{(n)}$ is latent, and the fixed τ_0^2, β sit outside the plate, shared across all n.



Question 2: Discrete Conjugacy

(a) Posterior. Write $\alpha_0 = \sum_{v=1}^{V} \alpha_v$ and $B(\boldsymbol{a}) = \prod_{v=1}^{V} \Gamma(a_v) / \Gamma\big(\sum_{v=1}^{V} a_v\big)$, so the prior density on the simplex $\Delta_{V-1} = \{\boldsymbol{\pi} : \pi_v \geq 0, \sum_v \pi_v = 1\}$ is $p(\boldsymbol{\pi}) = B(\boldsymbol{\alpha})^{-1} \prod_{v=1}^{V} \pi_v^{\alpha_v - 1}$; equivalently $\int_{\Delta_{V-1}} \prod_{v=1}^{V} \pi_v^{a_v - 1}\, d\boldsymbol{\pi} = B(\boldsymbol{a})$ for any $\boldsymbol{a}$ with positive entries, an identity reused in part (b). The likelihood is where the counts appear. Writing one categorical draw as $p(y_i \mid \boldsymbol{\pi}) = \pi_{y_i} = \prod_{v=1}^{V} \pi_v^{\mathbf{1}\{y_i = v\}}$ and using conditional independence given $\boldsymbol{\pi}$,

$$p(y_1, \ldots, y_n \mid \boldsymbol{\pi}) = \prod_{i=1}^{n} \prod_{v=1}^{V} \pi_v^{\mathbf{1}\{y_i = v\}} = \prod_{v=1}^{V} \pi_v^{\sum_{i=1}^{n} \mathbf{1}\{y_i = v\}} = \prod_{v=1}^{V} \pi_v^{n_v},$$

so the data enter only through the counts $(n_1, \ldots, n_V)$, a sufficient statistic, and the likelihood is a product of powers of the π_v, matching the functional form of the prior. That is exactly why the family is closed under conditioning. Multiplying,

$$p(\boldsymbol{\pi} \mid y_1, \ldots, y_n) \;\propto\; \prod_{v=1}^{V} \pi_v^{\alpha_v - 1} \prod_{v=1}^{V} \pi_v^{n_v} \;=\; \prod_{v=1}^{V} \pi_v^{(\alpha_v + n_v) - 1}.$$

The right side is the Dirichlet kernel with parameters $\alpha_v + n_v > 0$; a density is pinned down by its kernel, so the normalizer is $B(\boldsymbol{\alpha} + \boldsymbol{n})^{-1}$ and

$$\boxed{\boldsymbol{\pi} \mid y_1, \ldots, y_n \;\sim\; \text{Dirichlet}(\alpha_1 + n_1,\; \ldots,\; \alpha_V + n_V).}$$

The prior parameters behave as pseudocounts: the data simply add n_v to α_v, and the concentration grows from α_0 to $\alpha_0 + n$.

(b) Posterior predictive. Let $\boldsymbol{\alpha}' = \boldsymbol{\alpha} + \boldsymbol{n}$, so $\alpha'_0 = \sum_u \alpha'_u = \alpha_0 + n$. Given $\boldsymbol{\pi}$ the next draw is independent of the previous ones, so conditioning on $\boldsymbol{\pi}$ and marginalizing it out,

$$p(y_{n+1} = v \mid y_{1:n}) = \int_{\Delta_{V-1}} p(y_{n+1} = v \mid \boldsymbol{\pi})\, p(\boldsymbol{\pi} \mid y_{1:n})\, d\boldsymbol{\pi} = \int_{\Delta_{V-1}} \pi_v\, p(\boldsymbol{\pi} \mid y_{1:n})\, d\boldsymbol{\pi} = \mathbb{E}[\pi_v \mid y_{1:n}],$$

the posterior mean of π_v. The extra factor of π_v raises the v-th exponent by one, replacing $\boldsymbol{\alpha}'$ with $\boldsymbol{\alpha}' + \boldsymbol{e}_v$, so that same identity applies:

$$\mathbb{E}[\pi_v \mid y_{1:n}] = \frac{1}{B(\boldsymbol{\alpha}')} \int_{\Delta_{V-1}} \prod_{u=1}^{V} \pi_u^{(\alpha'_u + \mathbf{1}\{u=v\}) - 1}\, d\boldsymbol{\pi} = \frac{B(\boldsymbol{\alpha}' + \boldsymbol{e}_v)}{B(\boldsymbol{\alpha}')} = \frac{\Gamma(\alpha'_v + 1)}{\Gamma(\alpha'_v)} \cdot \frac{\Gamma(\alpha'_0)}{\Gamma(\alpha'_0 + 1)} = \frac{\alpha'_v}{\alpha'_0},$$

where the $u \neq v$ factors cancel and $\Gamma(z+1) = z\,\Gamma(z)$ is used twice. Hence

$$\boxed{p(y_{n+1} = v \mid y_1, \ldots, y_n) = \frac{\alpha_v + n_v}{\alpha_0 + n}, \qquad \alpha_0 = \sum_{u=1}^{V} \alpha_u.}$$

These sum to one over v, since $\sum_v (\alpha_v + n_v) = \alpha_0 + n$. Rewriting

$$\frac{\alpha_v + n_v}{\alpha_0 + n} = \frac{n}{\alpha_0 + n} \cdot \frac{n_v}{n} \;+\; \frac{\alpha_0}{\alpha_0 + n} \cdot \frac{\alpha_v}{\alpha_0}$$

exhibits it as a convex combination of the maximum likelihood estimate n_v/n and the prior mean α_v/α_0, with α_0 an effective prior sample size: it tends to the MLE as $n \to \infty$ but keeps positive mass on every category even when $n_v = 0$. This is additive smoothing, which becomes Laplace smoothing when $\alpha_v = 1$.

Question 3: Box's Loop

(1) Question. Runners describe "the wall" as an event – something that happens *to* you, at a place – and the physiology behind it is a phase transition: glycogen runs out after 90–120 minutes and the body switches to a slower fuel. If so, the field holds two kinds of second half, not one continuum. Is late-race deceleration generated by one continuous fatigue distribution or by a two-regime latent process? I treat this as model comparison with a guardrail: a second component must not merely fit better, it must *behave* like a physiological state.

(2) Data. The BAA publishes every finisher's net (chip) time at nine timing mats. I use the `llimllib/bostonmarathon` scrape of `baa.org` for 2013–2014: 48,148 finishers, cumulative splits at 5K, 10K, 20K, Half, 25K, 30K, 35K, 40K, finish, plus sex, age, bib. One row is one athlete; a missing mat is hardware, not a missing runner. (The loader also reads the 2015–2017 Kaggle release of the same feed, unreachable from this machine, so all numbers below are 2013–2014.) Hygiene: drop wheelchair/handcycle bibs and duplicates; read `"-"` and 0 as missing; blank any mat breaking strict monotonicity of cumulative time (a cross-read); repair a *single* interior gap by interpolation in distance, never a run of two, which would manufacture the smooth late trajectory I want to detect; drop implausible paces. Cost 153 rows, leaving 47,691 athletes whose four metric-defining mats are real reads.

$\log(\text{Half}_2/\text{Half}_1)$ is unusable at Boston: the course falls 140 m net from Hopkinton, most of it early, then climbs the Newton Hills (Fig. 1a). Gradient multiplies pace, so on the log scale the course is a common *constant* – harmless – but its size depends on ability, since slow runners pay more for the same hill (Fig. 1b). So I take a late window starting after the climb, a baseline ending before it, and subtract an ability-conditional reference:

$$y_i = \log \frac{p_i^{\text{late}}}{p_i^{\text{base}}} - \hat{\delta}_q(a_i), \quad p_i^{\text{late}} = \tfrac{1}{10}(T_{40} - T_{30}), \quad p_i^{\text{base}} = a_i = \tfrac{1}{15}(T_{25} - T_{10}),$$

$\hat{\delta}_q$ being the q=0.25 conditional quantile of the raw ratio over 40 ability bins. Within a stratum this is a pure location shift, so it *cannot* create bimodality; across strata it removes the ability-driven location differences that would manufacture it – conservative, that is, towards the hypothesis on trial.

(3) What the data cannot do. Survivorship: finishers only, and an athlete is likeliest to abandon in exactly the stretch y is measured over, so every π below is a lower bound and the censoring is informative; 2013 makes this visible, its "finishers" being those past the barricades when the race was stopped. *Truncation*: qualified entry makes this field a right tail of the recreational population. *Unobserved time-varying causes*: fuelling, cramping-versus-bonking and weather are invisible to a split file yet all three produce the same deceleration, so splits can show a second regime exists but not that glycogen defines it. Most valuable additions: DNF mat records, per-athlete fuelling and heart rate, years spanning a thermal contrast.

(4) Models. Three, hierarchical over g = sex × year, on the same y:

$$\mathcal{M}_0 : \; y_i \sim \mathcal{N}(\mu_g, \sigma_g^2); \qquad \mathcal{M}_s : \; y_i \sim \text{SkewNormal}(\xi_g, \omega_g, \alpha_g);$$

$$\mathcal{M}_2 : \; p(y_i \mid \theta) = (1 - \pi_g)\,\mathcal{N}(y_i \mid \mu_{0,g}, \sigma_{0,g}^2) + \pi_g\,\mathcal{N}(y_i \mid \mu_{1,g}, \sigma_{1,g}^2).$$

$\mathcal{M}_s$ is what makes the comparison mean anything: it buys the same right skew with one parameter instead of three, so if $\mathcal{M}_2$ wins only on skewness, $\mathcal{M}_s$ wins instead. I marginalise z_i out of $\mathcal{M}_2$ so NUTS sees a smooth posterior, and kill label switching structurally via $\mu_{1,g} = \mu_{0,g} + \exp(\lambda_g)$, enforcing the ordering on every draw while leaving λ_g free for its hierarchical prior. With $\sim 10^4$ athletes per group the group priors are *centred*: non-centring, the reflex choice, is what funnels when the likelihood is this strong. Responsibilities return by Rao-Blackwellisation, $P(z_i{=}1 \mid y_i) = \mathbb{E}_\theta[\pi_g \mathcal{N}_1 / ((1{-}\pi_g)\mathcal{N}_0 + \pi_g \mathcal{N}_1)]$.

(5) Conclusion: two components, but not the two I was promised. PSIS-LOO prefers $\mathcal{M}_2$ over $\mathcal{M}_s$ by $\Delta\text{ELPD} = 1275.0 \pm 128.2$ (9.9 s.e.) and over $\mathcal{M}_0$ by 5667.8 ± 163.8, with $p_{\text{loo}} = 31.8$, every Pareto-k below 0.41, and visibly flatter LOO-PIT (Fig. 1e): the extra mass is not merely skew. Three checks qualify that.

(a) Refit to draws from a genuinely unimodal skew-normal, the same pair puts **??** on top: the test does not just reward flexibility. *(b)* 2013–2014 has no thermal contrast, so this is calibration, not evidence: planting π = **??**% in one year against $\approx$ 19% in two others and grouping only by year recovers it within **??** points (**??** intervals cover truth), with no weather covariate anywhere. *(c)* The model fails this one. Matched on baseline pace, the standardised difference over the first two 5K segments is -0.48 and -0.69 – negative, so the wall class ran them *faster* – while the shape *within* the first 10K is flat (-0.06): they went out hard and were already fading inside the baseline window, so y understates them. And since the wall component is the wider one, its responsibility exceeds 1/2 in *both* tails, flagging large negative splits ($y < -0.143$) beside true collapses ($y > 0.162$). Against Smyth (2021), who calls a wall at a 20% slowdown from 5K–20K to 25K–finish and finds 28% of men and 17% of women over 4.1M races: his rule run verbatim here flags 33% and 23% – Boston's hills inflate a fixed threshold – yet the de-confounded mixture independently recovers 28% and 17%. Same incidence, not the same athletes, though: Cohen's κ is only 0.57 and 0.44, and at equal costs the Bayes rule calls 19% and 9%, needing false negatives weighted 2.7× and 3.7× to reach his.

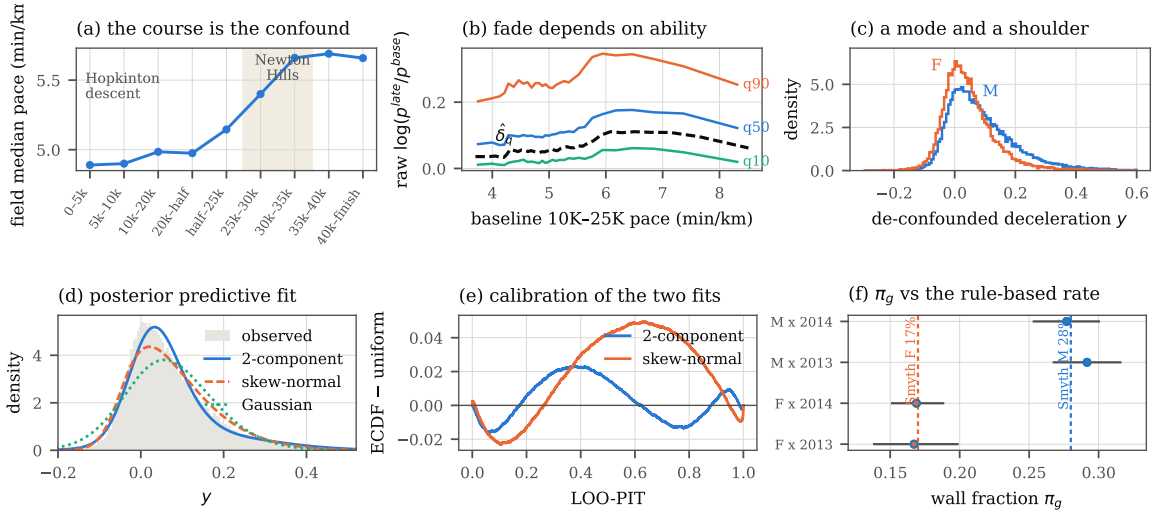


Figure 1: (a) Field median pace per segment: the descent makes the opening cheap and the Newton Hills (shaded) make the approach dear, so a "second half" statistic prices terrain as fatigue. (b) Raw-ratio quantiles against ability; dashed is the reference $\hat{\delta}_q$ – far from flat, so the course effect is not one constant. (c) The de-confounded metric y by sex. (d) Posterior predictive densities: the Gaussian misses peak and shoulder, the skew-normal the shoulder only. (e) LOO-PIT departure from uniformity. (f) Posterior π_g with 94% intervals, against Smyth's incidence.

(6) Next step: promote the label to a location. A scalar cannot say *when* deceleration began, and depletion is a claim about when. Over all nine splits, with d_s the midpoint of segment s, ℓ_{is} log pace, α_i the athlete's level, δ_s a sum-to-zero course profile shared by the field, κ_i the change-point, $\beta_i > 0$ the fade rate and $m_{is} = \alpha_i + \delta_s + \beta_i (d_s - \kappa_i)_+$:

$$p(\ell_{i\cdot}) = (1 - \pi_g) \prod_s \mathcal{N}\big(\ell_{is} \mid \alpha_i + \delta_s,\, \sigma^2\big) + \pi_g \prod_s \mathcal{N}\big(\ell_{is} \mid m_{is},\, \sigma^2\big).$$

This repairs all three defects at once: δ_s puts the topographic correction *inside* the model, $\beta_i > 0$ makes the wall directional so no negative split can be flagged, and κ_i makes the hypothesis falsifiable – depletion predicts κ massed near 30–35 km, and heat moving it earlier as well as raising π. On a subsample the median change-point is already **??** km (10–90%: **??**–**??**); refitting on 2015–2017 with DNF records restored would settle the question.